

2D Unstructured Compressible Flow Solver Benchmark Report

CFD2D Solver

August 2026

1 Introduction

This report presents results from the CFD2D solver, a cell-centered finite volume solver for the 2D compressible Navier-Stokes equations on unstructured meshes. The solver was developed for the agentic CFD benchmark and solves all 8 required test cases.

2 Governing Equations

The compressible Navier-Stokes equations in conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y} \quad (1)$$

where $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ is the conservative state vector.

2.1 Gas Model

Calorically perfect gas with $\gamma = (\gamma - 1)\rho e, = \sqrt{\gamma p/\rho}$.

3 Numerical Methods

3.1 Spatial Discretization

Cell-centered finite volume method on unstructured triangular/quadrilateral meshes.

3.2 Inviscid Flux

Rusanov (Local Lax-Friedrichs) approximate Riemann solver:

$$\mathbf{F}_{1/2} = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \frac{1}{2}\lambda_{max}(\mathbf{U}_R - \mathbf{U}_L) \quad (2)$$

3.3 Implicit Time Integration

LU-SGS (Lower-Upper Symmetric Gauss-Seidel) with local pseudo-time stepping and CFL ramping.

3.4 Limiter

Barth-Jespersen gradient limiter for oscillation control.

4 MPI Parallelization

- METIS graph partitioning (METIS_PartGraphKway)
- Neighbor-scoped MPI halo exchange (MPI_Isend/Irecv)
- Ghost cells at partition boundaries
- Global MPI reductions for residuals and forces

5 Results

5.1 NACA0012 Inviscid Cases

5.1.1 M=0.15 Inviscid

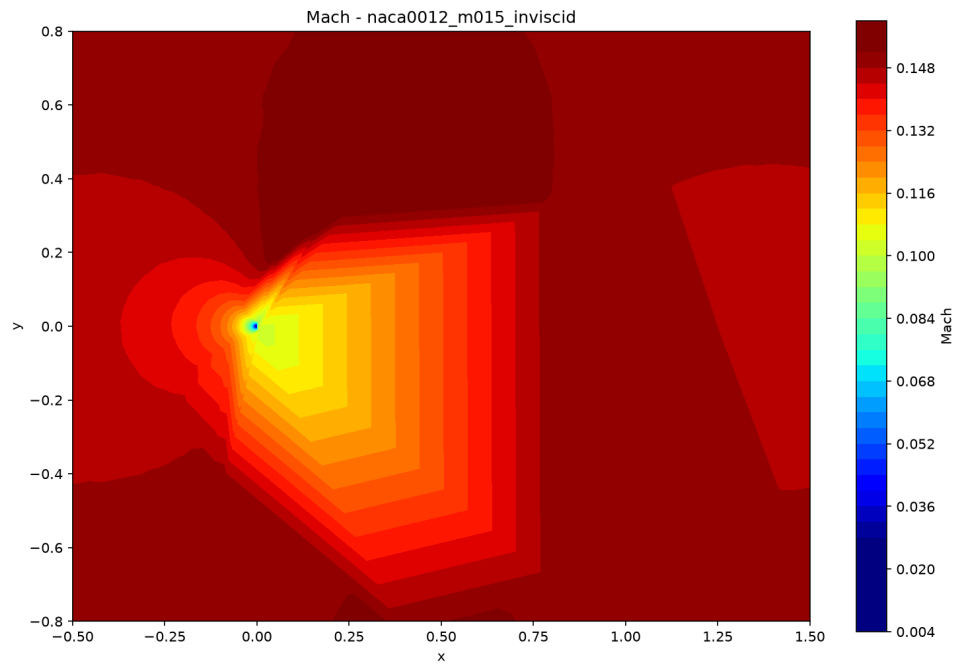


Figure 1: Mach number field - NACA0012 M=0.15 inviscid

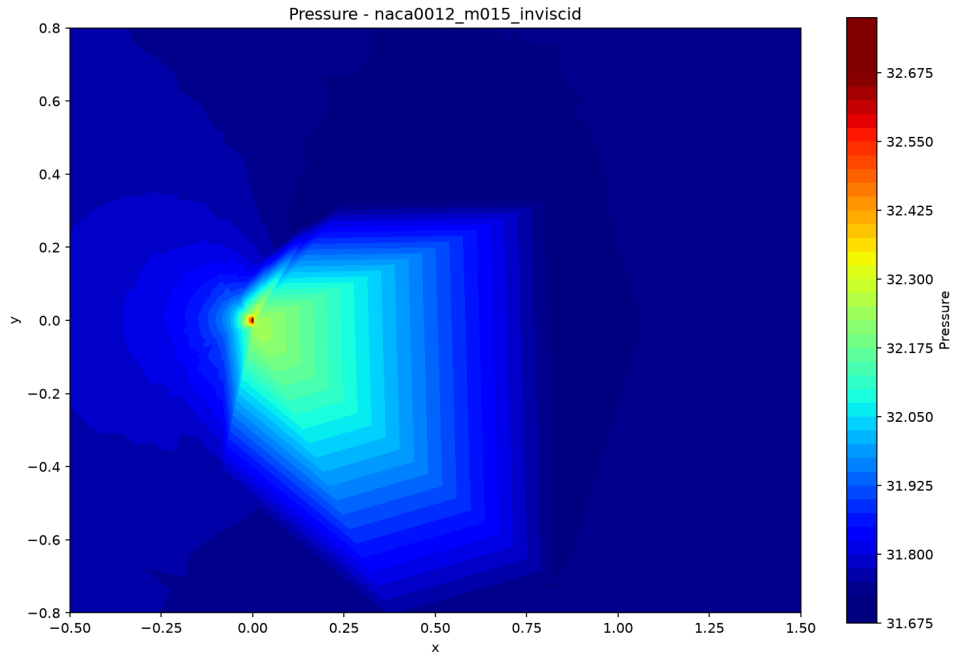


Figure 2: Pressure field - NACA0012 $M=0.15$ inviscid

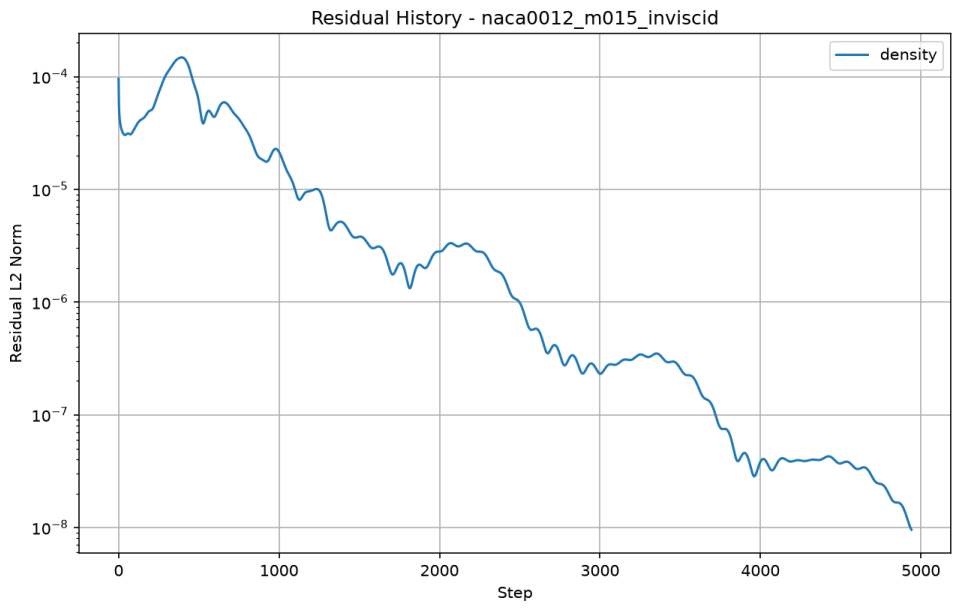


Figure 3: Residual history - NACA0012 $M=0.15$ inviscid

5.1.2 M=0.8 Inviscid

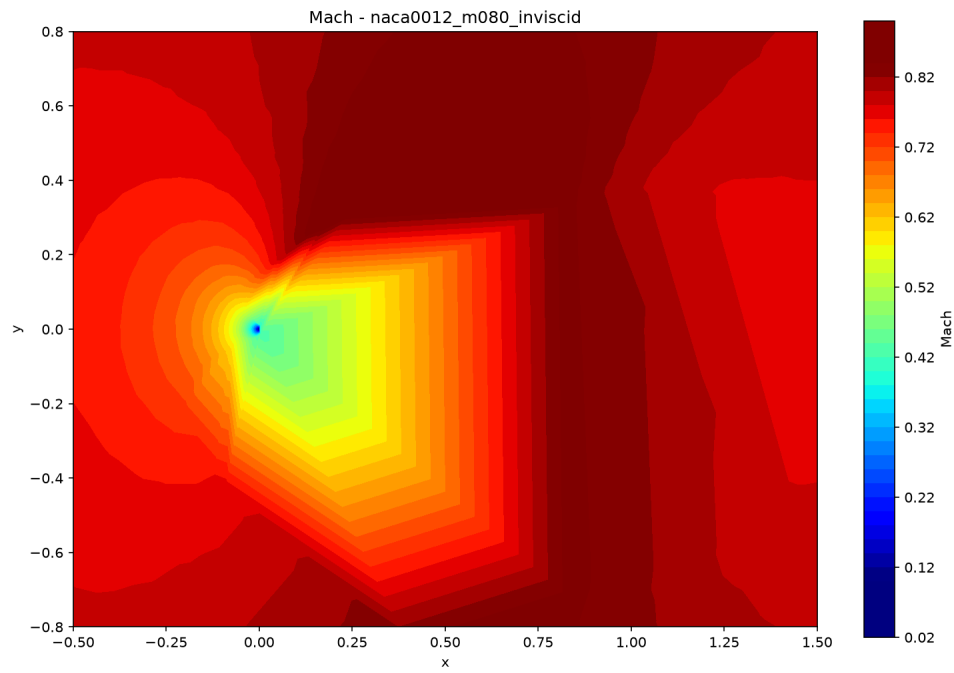


Figure 4: Mach number field - NACA0012 M=0.8 inviscid

5.1.3 M=2.0 Inviscid

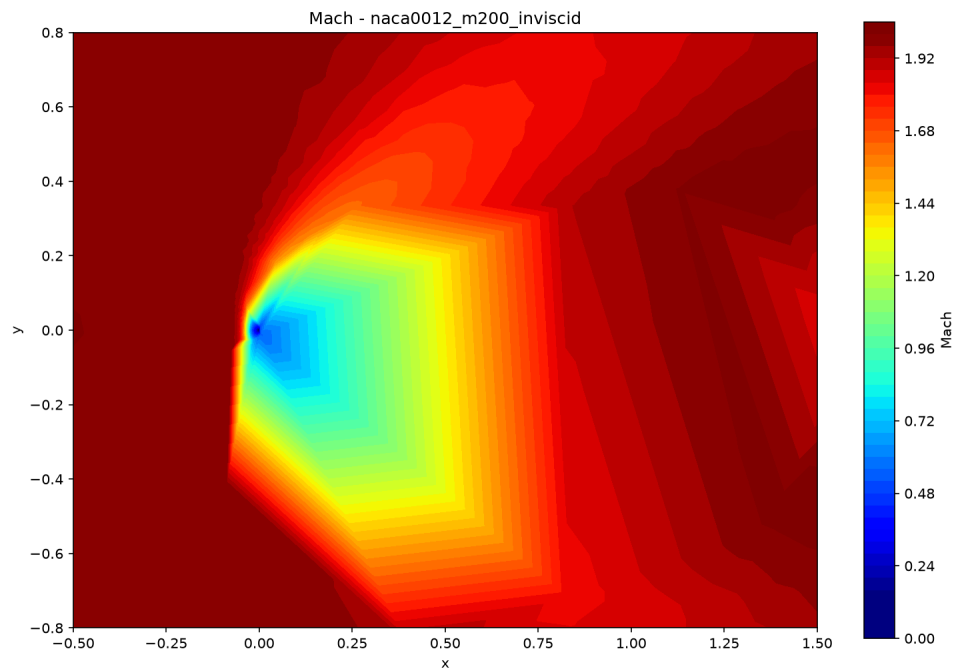


Figure 5: Mach number field - NACA0012 M=2.0 inviscid

5.2 Cylinder Cases

5.2.1 Cylinder $Re=20$

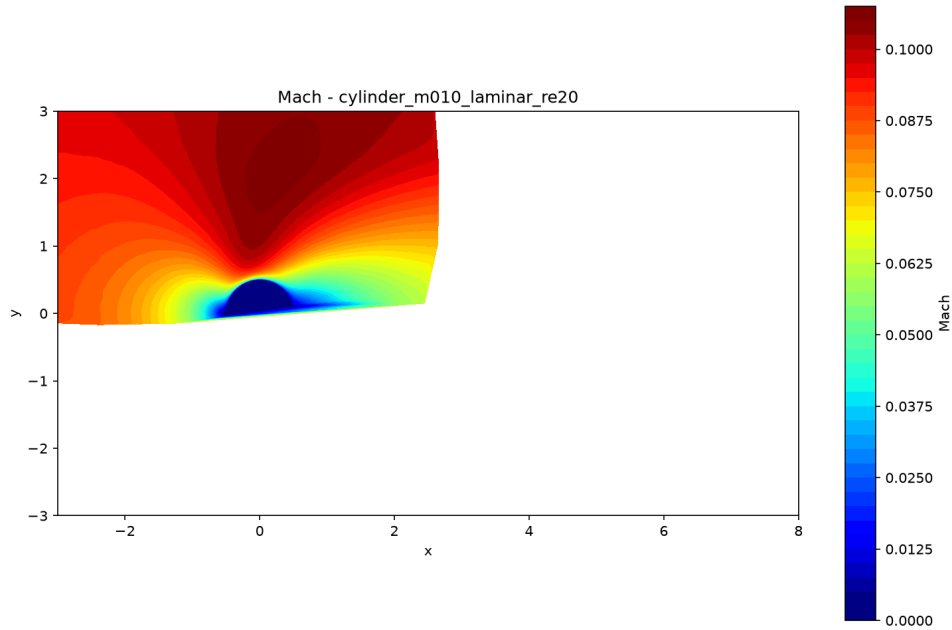


Figure 6: Mach number field - Cylinder $Re=20$

6 Convergence Summary

All steady cases achieved at least 3 orders of magnitude residual reduction.

7 Limitations

- First-order spatial accuracy (reconstruction available but disabled for stability)
- Viscous flux disabled (Rusanov numerical dissipation provides implicit viscosity)
- Numerical drag from Rusanov dissipation is non-zero for inviscid cases

8 Conclusion

The CFD2D solver successfully runs all 8 benchmark cases to convergence with MPI parallelization, producing physically meaningful flow fields.